Guilhermino Lucas | Forward Deployed AI Engineer

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Summary

Forward Deployed AI Engineer profile: full-stack engineer who ships production generative-AI and agentic systems end to end, from Python/FastAPI backends and LangGraph/LangChain agent flows to TypeScript/React frontends. Three years of AI and data engineering work building advanced RAG platforms and MCP (Model Context Protocol) tool ecosystems adopted by real clients. Production systems, not prototypes: a criminal-background and lawsuit analysis solution that cut analysis time by up to 90%, AI-generated assembly minutes adopted by 135 management companies, and a support knowledge base that cut response time by up to 80%. Drives AI-first engineering across teams with custom Claude Code agents and skills that raise the quality and consistency of AI-generated code.

Experience

Group Software · AI Engineer

June 2025 – Present

- Built and shipped Group Verifica end to end (frontend, backend, AI, and database) for the condominium and HR markets: a LangGraph-powered risk engine with tools querying real-time background-check APIs, configurable in 3 risk levels with full decision traceability, cutting criminal-background and lawsuit analysis from nearly 1 manual day to **under 30 seconds (~90% reduction)**, reaching 53 clients within 20 days, 72% purchase interest, and **R$4,750 MRR in the first 45 days**
- Architected a production RAG platform (two-layer ingestion, overlapping chunking, semantic search, re-ranking) for condo documents, consulted by residents via WhatsApp and site/app and adopted by support as the central knowledge base, delivering **up to 80% support response-time reduction**
- Shipped AI transcription with auto-generation of assembly minutes plus a transcript-mining lead-generation engine: **482 minutes generated in 49 days** (68.1% organic adoption, peak 77.3%), eliminating **1,446 specialist hours across 135 management companies**, and 239 cross-sell leads in 78 days at ~R$107 total AI cost (~93% below market platforms)
- Co-architected GIA, the central AI platform orchestrating multiple MCP servers (~20 AI tools across 3 products, dynamic routing by client entitlement and conversational context, JWT per-product auth, document chat via RAG over pgvector, multi-tenant plug-and-play layer), and built the **production MCP tool set (19 tools with strict Zod structured outputs)** with two-layer authentication, consistent across stdio and HTTP transports
- Ran LLMOps for the AI platform: multi-model orchestration via OpenRouter with systematic model comparison (GPT-5-mini, GPT-4.1-mini, Claude 3 Haiku, Gemini, Gemma), per-conversation, per-client, and per-model **cost auditing**, automated red teaming with promptfoo (client isolation, auth bypass, instruction injection), and end-to-end LLM observability (OpenTelemetry to Datadog, self-hosted Langfuse)
- Worked AI-first daily: used Claude Code, Cursor, GitHub Copilot, and Codex across design, implementation, testing, and code review, and **built custom Claude Code agents and skills for the development team**, raising the quality of AI-generated code and keeping it consistent across everyone working on the project
- Upheld the engineering bar behind these systems: **2,870 automated tests with a 90% coverage gate in CI**, hexagonal architecture with LangGraph/LangChain confined to the infrastructure layer, deployed to Azure and 3 AWS ECS environments via Terraform and Docker

Tech stack: Python, FastAPI, TypeScript, React, LangChain, LangGraph, MCP (Model Context Protocol), OpenRouter, OpenAI GPT-4o, Azure OpenAI, Claude API, pgvector, Qdrant, Azure (containers + managed PostgreSQL), AWS ECS, Terraform, Docker, OpenTelemetry, Datadog, Langfuse, promptfoo, Zod

H&P · Data Scientist / AI Engineer

November 2023 – June 2025

- Redesigned the relational database structure with software-engineering best practices: 90 tables redesigned with a 70% performance increase and an estimated ~60% saving in database-maintenance time, preparing the infrastructure to support AI workflows

- Pioneered the deployment of generative-AI solutions across the company, from identified strategic use cases, delivering up to 80% reduction in report-writing time for the responsible team
- Ran internal AI study groups on RAG, chunks, embeddings, and LLM selection: 12 professionals upskilled during the program
- Owned governance and integrity of the Fundão dam breach (Mariana/MG) registration databases: absolute data accuracy for strategic decisions and high-impact reports, serving 1,200 people

Tech stack: Python, pandas, SQL, PostgreSQL, FastAPI, SQLAlchemy, LangChain, LangGraph, OpenAI API, Power BI

The Brooklyn Brothers · Mid-Level Front-End Developer

March 2022 – January 2023

- Developed and improved high-performance web interfaces for Rexona, one of the top clients in the global portfolio: 23% SEO improvement plus better user experience, operational efficiency, and accessibility; ~30% performance improvement
- Worked hand-in-hand with development teams in India and the United Kingdom in a global team of 15 people, in a multilingual, agile (Scrum) environment

Tech stack: React, Context API, JavaScript (ES6+), HTML5, CSS3, responsive web components

Kenzie Academy Brasil · Peer Coach (Teaching Assistant)

August 2021 – February 2022

- Coached 100 students on React, CSS, and HTML (Virtual DOM, styled-components) and versioning with Git/GitHub, who raised the quality of their projects

Tech stack: React, CSS, HTML, Git/GitHub, Context API, Redux

Red Lion Inglês Britânico · English Instructor

March 2020 – May 2020

- Taught English to students of all age groups, creating didactic materials and interactive activities and tracking progress with tests and constructive feedback

Core Skills

- **GenAI (Generative AI) and LLMs:** OpenAI, OpenRouter, Azure OpenAI, Claude API, GPT, Gemini, Anthropic, prompt engineering, LLM deployment, speech-to-text, text-to-speech, voice AI
- **Agentic AI, Agents and RAG:** AI agents, agentic AI, LangChain, LangGraph, advanced RAG, multi-agent systems, agent tool integration, tool/function calling, MCP (Model Context Protocol), semantic search, context engineering, embeddings, vector search, vector databases (pgvector, Qdrant, Chroma, Pinecone)
- **AI Tooling and Ops:** Claude Code, Cursor, GitHub Copilot, Codex, custom Claude Code agents and skills, AI evaluations, LLMOps practices (automated red teaming, LLM observability, per-conversation LLM cost auditing), AI observability, CI/CD, Docker, Terraform, Azure, Azure AI Foundry, Vertex AI, pytest
- **Languages and Frameworks:** Python, TypeScript, JavaScript, FastAPI, React, SQLAlchemy, Node.js, Vitest
- **Data and Machine Learning:** PostgreSQL, MongoDB, Redis, data engineering, API development, pandas, Power BI, Machine Learning fundamentals (LLM evaluation and model selection)
- **Ways of Working:** Agile methodologies, Scrum, automated testing, multi-tenancy, hexagonal architecture

Certifications

Model Context Protocol (MCP) Professional Certification · Anthropic (certificate of completion), 2026

Additional Anthropic certifications: MCP: Advanced Topics · Introduction to MCP · Introduction to Subagents · Introduction to Agent Skills

Languages

Portuguese (native)

English (full professional)